Topology Under-Determines Governance Maturity in Data Lineage
Graphs: A Negative Result and a Reusable Inference Protocol

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Abstract

A recurring assumption in data-governance tooling and in the lineage-graph literature is that governance maturity leaves a measurable structural signature: better-governed estates should produce topologically distinct lineage graphs. We test that assumption directly. On a curated-metadata production dbt estate (18 analyzable domains), a spectral-gap descriptor correlates strongly with documentation rate (Spearman $\rho = -0.71$, bootstrap 95% CI $[-0.92, -0.27]$). The correlation does not mean what it appears to: the descriptor takes three distinct values across the eighteen domains, a layer-stratified permutation test returns $p = 1.000$, and the relationship is carried entirely by between-layer architecture (source, staging, mart), not within-layer governance. A rank-degeneracy power analysis shows the observed effect sits at the design's minimum detectable size. The relationship does not replicate across organizations, and the apparent non-replication is confounded with a domain-construct mismatch we make explicit. A persistent-homology descriptor reduces exactly to the cycle-rank baseline ($\rho = +1.000$) on real graphs, and a node-importance prediction result (AUC 0.90) reproduces a published centrality finding rather than extending it. Our contribution is therefore twofold and deliberately deflationary: (1) a carefully bounded negative result, that lineage topology is a poor proxy for governance maturity once the architectural-layer confound is removed; and (2) a reusable inference protocol for small-n graph-descriptor correlation studies, covering permutation and FDR control, layer-stratified permutation, degree-preserving nulls, and a rank-degeneracy-aware power analysis, packaged as tested, runnable code. The protocol is the transferable artifact: it lets future structure-versus-attribute studies distinguish a real relationship from one manufactured by rank degeneracy or a confounding partition.

Keywords: data governance, lineage graphs, community detection, spectral graph theory, persistent homology, structural descriptors, network analysis

1 Introduction

Data governance frameworks specify how organizations should manage data assets: who owns them, how quality is assured, what documentation exists, and how changes propagate through transformation pipelines. Mature governance manifests as clear domain boundaries, tested transformations, and documented interfaces between stages of a data pipeline. Immature governance manifests as orphaned assets, unclear ownership, redundant paths, and untested transformations.

The dominant approach to assessing governance maturity relies on self-assessment against capability models such as DAMA-DMBOK [DAMA International, 2017] or DCAM. These frameworks are reference guides: they organize knowledge areas and suggest best practices, but do not prescribe a single maturity ladder. More fundamentally, they capture stated policy rather than the actual structural organization of data assets. Two organizations with similar self-assessment scores can have very different lineage topologies: one with clean layered DAGs and well-separated domain boundaries, another with tangled cross-domain dependencies and redundant paths.

We hypothesize that governance practices leave structural fingerprints in data lineage graphs. If domain boundaries are enforced, community detection at the appropriate resolution should recover stable, modular partitions. If transformations are tested, the blast radius of any single failure should be concentrated rather than diffuse. If the pipeline architecture is clean, the spectral properties of the lineage graph should reflect appropriate coupling. If redundant dependency paths exist (a sign of organic growth without architectural oversight), persistent homology should detect reconvergent structure in the undirected skeleton.

This paper introduces four multi-scale structural descriptors designed to capture these properties, validates them on synthetic lineage graphs with controlled governance variation, tests their sensitivity to governance-relevant interventions, and provides initial validation on a real production dbt lineage graph with genuine governance metadata.

1.1 Contributions

1. **Cross-dataset structural characterization**, the paper's primary empirical contribution. Four graph-theoretic descriptors (D1–D4) are computed on 32 graphs from four heterogeneous sources (WfCommons scientific workflows, DLG-DG-23 Huawei Cloud lineage, DW-Bench warehouse schemas, anonymized production dbt manifest). Principal component analysis shows that production data lineage graphs occupy a distinct descriptor region characterized by high community stability and small spectral gaps. This is reproducible, descriptive, and does not depend on any governance metadata.

2. **Precise descriptor definitions and implementation**, including LWCC handling for disconnected graphs, combinatorial (unnormalized) Laplacian for D3, GUDHI parameters for D4 (`max_dimension=2`, $\mathbb{F}_2$ coefficients), and cycle rank baseline. All computations are reproducible from the public Zenodo archive.

3. **Methodological scaffolding for governance–topology studies**: permutation-based inference, Benjamini–Hochberg FDR correction with explicit hypothesis families, rank-residualization partial correlations, bootstrap confidence intervals, two null models (degree-preserving and layer-preserving), subset robustness, layer-stratified permutation, D1 seed-robustness, leave-scale-out cross-validation, and power analysis under the rank-degenerate structure observed in real data.

4. **A single-organisation pilot of governance–topology correlation** on a production dbt lineage with explicit acknowledgement of rank-degeneracy in D3 (3 effective values across 18 domains), detection-threshold-level effect size (observed $|\rho| = 0.71$ vs. minimum detectable ≈ 0.7–0.8 at 80% power), and layer-stratified permutation $p = 1.0$ showing the association is between-layer architectural rather than within-layer governance signal.

5. **Negative cross-organisation pilot at graph level with methodological caveat.** D3/documentation does not replicate to two public dbt projects (Cal-ITP, Mattermost) under folder-derived domain partitioning. We discuss why folder-derived domains are not the same construct as metadata-assigned dbt domains.

6. **A reproduction of a published centrality result at node granularity.** On the DLG-DG-23 dataset with expert-annotated core data assets (36 labels across 6 Huawei Cloud graphs from Table 5 of [Chen et al., 2024]), node-level topological descriptors trained

on five graphs predict core-asset status in the held-out sixth graph at mean LR AUC $= 0.898 \pm 0.098$ (LOGO cross-validation), substantially above random (0.546) and out-degree-only (0.576) baselines. This reproduces the node-centrality finding of Chen et al. [2023] on the same corpus: the top predictors are PageRank and out-degree. Phase 5 evaluates ordinary node-level centrality, reachability, degree, and clustering features; it does not evaluate incremental D1–D4 information. We report it as a reproduction, not as a new governance result.

2 Related Work

2.1 Governance Measurement

Data governance frameworks define structural and procedural mechanisms for data management [Khatri and Brown, 2010, Abraham et al., 2019]. Assessment typically relies on maturity models: DAMA-DMBOK [DAMA International, 2017] organizes eleven knowledge areas as a reference guide (not a prescriptive maturity ladder); DCAM evaluates institutional readiness. Weber et al. [2009] argue that governance design is contingent on organizational strategy, implying that a single maturity scale is insufficient.

Institutional theory provides a theoretical lens. DiMaggio and Powell [1983] describe three isomorphic pressures (coercive, mimetic, normative) that cause organizations to converge structurally under shared governance regimes. Scott [2013] extends this via regulative, normative, and cultural-cognitive pillars. We adopt this view: if governance is an institutional arrangement, it should produce detectable structural signatures in the artifacts it governs.

2.2 Graph Analysis of Data Pipelines

Hou et al. [2026] apply temporal convolutional and graph convolutional networks to data lineage DAGs for fault classification (1,240 nodes, 5,870 edges), achieving 95.8% accuracy. Their work uses lineage structure as input to prediction but does not extract governance-relevant patterns from that structure.

Knowledge graph quality metrics [Seo et al., 2022] define six structural measures for knowledge graphs. Meroño Peñuela et al. [2024] position knowledge graphs as governance backbones for AI systems. Neither applies multi-scale graph descriptors to lineage topology.

An unpublished phase 0 feasibility analysis by the author [Malempati Hari, 2026b] asked whether box-covering fractal dimension could characterize a governance signal in data lineage. It could not. Box-covering needs a diameter of roughly 8 or more and a locally connected lattice-like topology before its scaling estimates stabilize, and enterprise governance graphs supply neither. Lineage DAGs run to a diameter of 3–6, entity-relationship graphs to 2–4 over a bipartite hub topology, and product hierarchies are trees on which box-covering is trivial. We read that as the geometry of the domain rather than a deficiency in the data, since a well-governed lineage is a shallow DAG by design. The descriptors used here were chosen to fit that geometry, and we make no fractal claim about governance graphs anywhere in this paper. The fractal analysis methodology (Hurst exponents, detrended fluctuation analysis) has a longer history in the author's work on temporal scaling in financial systems [Malempati Hari, 2026c]. A complementary line of work applies boundary-risk detection to master data governance [Malempati Hari, 2026a], identifying entity clusters whose governance status is sensitive to local threshold perturbation. That approach targets record-level entity resolution rather than pipeline-level lineage topology, but shares the premise that governance properties are recoverable from structural analysis of data artifacts.

2.3 Prior Findings This Work Confronts

Three published results bound what we can and cannot claim, and we position our experiments against them explicitly rather than as novel discovery.

Chen et al. [2023] show that node-centrality metrics identify core data assets on the DLG-DG-23 lineage graphs. Our Experiment 7 (core-asset prediction, AUC 0.90) is, under leave-one-graph-out cross-validation on the same corpus, a reproduction of that result: the top predictors are PageRank and out-degree. The experiment contains ordinary node features and does not test D1–D4 at node granularity. We therefore make no claim about their incremental value from this analysis. We report the AUC as confirmation that lineage structure carries node-importance signal already captured by simple centrality, not as a new finding and not as evidence that multi-scale descriptors detect governance.

The DLG-DG-23 dataset paper [Chen et al., 2024] already characterizes data-lineage graphs as sparse, scale-free, and cluster-rich. We therefore scope our cross-source structural comparison (Experiment 5) narrowly, as a contrast between production lineage, scientific workflows (WfCommons), and warehouse schemas (DW-Bench), rather than as new structural discovery.

Spectral change-point detection on dynamic graphs is a solved problem with a published, benchmarked method: LAD [Huang et al., 2020] and its multi-view extension MultiLAD [Huang et al., 2024]. Our longitudinal descriptor-monitoring idea is an under-powered re-derivation of it; we do not claim change detection as a contribution, and we defer the git-history dbt time series to a separate dataset/resource paper where the contribution is the data, not the detector.

2.4 Community Detection at Multiple Resolutions

The resolution parameter in modularity optimization [Reichardt and Bornholdt, 2006] controls community granularity. Lambiotte et al. [2014] interpret the parameter as Markov diffusion time. Delvenne et al. [2010] introduce stability as plateau detection via normalized variation of information (NVI). The Leiden algorithm [Traag et al., 2019] guarantees connected communities, addressing a known Louvain failure mode. Our implementation uses the Louvain algorithm (via NetworkX) rather than Leiden, as the resolution sweep measures partition stability across resolutions rather than depending on within-resolution community connectedness guarantees.

2.5 Spectral Methods

Algebraic connectivity (the second-smallest Laplacian eigenvalue) measures graph cohesion [Fiedler, 1973]. Spectral gap correlates with robustness to random failure and inversely with mean path length. Von Neumann spectral entropy [De Domenico and Biamonte, 2016] quantifies complexity of the eigenvalue distribution. We compose these into a descriptor that captures coupling structure between pipeline components.

2.6 Persistent Homology on Networks

Topological data analysis via persistent homology has been applied to brain networks, social networks, and protein interaction networks [Otter et al., 2017, Petri et al., 2013]. Chowdhury and Mémoli [2018] introduce directed persistent path homology. For DAGs, directed H1 is trivially zero (no directed cycles by definition), but H1 in the undirected skeleton detects reconvergent dependency paths where multiple routes exist between a source and its consumers. To our knowledge, no prior work applies persistent homology to data lineage or governance graphs.

3 Descriptors

We define four structural descriptors for directed acyclic data lineage graphs $G = (V, E)$ with $|V| = N$ nodes and $|E| = M$ edges. Each node represents a data asset (source, transformation, or mart). Edges represent dependency: $(u, v) \in E$ means v depends on u.

Graph preprocessing. Production lineage graphs are often weakly disconnected: isolated source tables and small clusters that share no edges with the main pipeline appear in every real dataset we examined. Descriptors D1, D3, and D4 are computed on the largest weakly connected component (LWCC) of the undirected projection of G, since community detection and spectral methods require a connected graph. D2 (blast-radius) uses the full directed graph because downstream reachability is defined per node regardless of global connectivity. For the production dbt graph, the LWCC contains 185 of 223 nodes (83%); the 38 remaining nodes are source tables outside the main pipeline (36 isolated singletons plus one disconnected pair, which carries the single edge dropped when restricting to the LWCC, 263 $\rightarrow$ 262).

3.1 D1: Community Stability Index (CSI)

We sweep the resolution parameter γ of the Louvain algorithm over $[0.1, 10.0]$ in 20 logarithmically spaced steps on the undirected skeleton of G, computing community partitions $\mathcal{C}(\gamma)$ at each resolution. For adjacent resolution pairs (γ_i, γ_{i+1}), we compute the normalized variation of information:

$$\mathrm{NVI}(\gamma_i, \gamma_{i+1}) = \frac{\mathrm{VI}(\mathcal{C}_i, \mathcal{C}_{i+1})}{\log_2 N} \tag{1}$$

where $\mathrm{VI} = H(\mathcal{C}_i|\mathcal{C}_{i+1}) + H(\mathcal{C}_{i+1}|\mathcal{C}_i)$ is the variation of information and $\log_2 N$ is the maximum possible entropy for N nodes. The Community Stability Index counts the fraction of consecutive resolution steps where the partition remains stable:

$$\mathrm{CSI} = \frac{|\{i : \mathrm{NVI}(\gamma_i, \gamma_{i+1}) < \tau\}|}{n_{\mathrm{steps}} - 1}, \quad \tau = 0.1 \tag{2}$$

$\mathrm{CSI} \in [0, 1]$, with higher values indicating more stable community structure across resolutions. We additionally extract fragmentation onset (the γ at which the number of communities first exceeds twice the minimum), modularity at $\gamma = 1$, and number of communities at $\gamma = 1$.

Louvain stochasticity, and a second variance source the audit does not cover. The Louvain algorithm is a heuristic with random seed dependence. We audited D1 CSI over 25 random seeds per graph and report seed-NVI at $\gamma \approx 1.0$ (mean pairwise NVI between seed partitions at a fixed resolution) as a measure of optimizer variance. $\mathrm{CSI}_{\mathrm{std}} < 0.05$ and seed-NVI < 0.05 across those seeds.

That audit holds the graph fixed and varies only the optimizer's seed, and it does not bound the variance that matters most here. Louvain is sensitive to the order in which nodes are inserted, and our extraction built each graph from an unordered Python set, so insertion order followed the interpreter's per-process string hash randomization rather than anything about the data. Holding one graph fixed and permuting only the insertion order, D1 CSI takes values spanning 0.500 to 0.786, a range of 0.286 against a corpus standard deviation of 0.123. The optimizer-seed audit is therefore not evidence that D1 is reproducible, and the earlier claim to that effect does not follow from it. Insertion order is now sorted and the hash seed pinned, which makes D1 deterministic going forward but not comparable to previously released values. D2 and D4 are unaffected, and D3 algebraic connectivity and normalized gap reproduce to 10^{-14}; D3 Fiedler bimodality varies at 5×10^{-6} relative, since the Fiedler vector is not unique under near-degenerate eigenvalues.

Governance interpretation. Well-governed pipelines exhibit high CSI: communities are robust across resolutions because they reflect genuine organizational structure. Poorly governed pipelines with tangled cross-domain dependencies exhibit low CSI.

3.2 D2: Blast-Radius Concentration Profile

For each node $v \in V$ and depth $d \in \{1, 2, \ldots\}$, we compute $R_d(v) = |\{u : \mathrm{dist}(v, u) = d\}|$, the number of nodes reachable at exactly depth d downstream. The Gini coefficient of $\{R_d(v)\}_{v \in V}$ at each depth d yields a concentration profile:

$$G(d) = \mathrm{Gini}(\{R_d(v) : v \in V\}) \tag{3}$$

We extract the maximum Gini across depths and the concentration transition depth (the depth at which Gini first exceeds 0.5, or -1 if never reached).

Governance interpretation. In well-governed pipelines, blast radius should be concentrated: a few hub nodes affect many downstream assets, while most nodes have limited impact. High max Gini indicates concentrated risk, which, when combined with stewardship data, can reflect intentional architectural design.

3.3 D3: Spectral Gap and Fiedler Structure

From the combinatorial (unnormalized) Laplacian $L = D - A$ of the undirected skeleton, where D is the degree matrix and A the adjacency matrix, we extract:

- Algebraic connectivity λ_2: the second-smallest eigenvalue of L on the LWCC. Because the LWCC is connected by construction, $\lambda_1 = 0$ and $\lambda_2 > 0$ always.
- Normalized spectral gap: $\lambda_2/\lambda_{\max}$, where $\lambda_{\max}$ is the largest eigenvalue.
- Fiedler bimodality: bimodality coefficient of the Fiedler vector (the eigenvector associated with λ_2).
- Spectral entropy: von Neumann entropy of the eigenvalue distribution, $S = -\sum_i \bar{\lambda}_i \log_2 \bar{\lambda}_i$ where $\bar{\lambda}_i = \lambda_i / \sum_j \lambda_j$ (computed over nonzero eigenvalues).

Governance interpretation. Algebraic connectivity measures overall graph cohesion. Low values indicate loosely connected components (possibly orphaned domains). High Fiedler bimodality indicates a natural bipartition, potentially reflecting a two-tier architecture (e.g., source vs. mart layers). Spectral entropy captures the complexity of the coupling structure.

3.4 D4: Persistent Homology via Shortest-Path Filtration

We compute all-pairs shortest-path distances on the LWCC of the undirected skeleton and construct a Vietoris–Rips complex using GUDHI [Otter et al., 2017] with `max_dimension=2` (so that the simplex tree includes 2-simplices, enabling H1 deaths) and `max_edge_length` set to the graph diameter plus one. Persistence is computed over $\mathbb{F}_2$ coefficients; zero-persistence intervals (birth = death) and infinite bars are discarded. Domain subgraphs with $N < 3$ or no internal edges contribute zero H1 bars by convention. For each filtration threshold ϵ, we compute homology in dimensions 0 and 1, and record persistence diagrams of birth–death pairs.

From the H1 (cycle) persistence diagram, we extract:

- Number of H1 bars (n_{H1}) and its normalization n_{H1}/N.
- Total H1 persistence: $P_1 = \sum_i (d_i - b_i)$ over all H1 bars.
- H1 persistence entropy: $S_1 = -\sum_i p_i \log_2 p_i$ where $p_i = (d_i - b_i)/P_1$.

As a simpler baseline, we also compute the cycle rank $\beta_1 = M - N + C$ of the undirected skeleton (where C is the number of connected components), which counts the total number of independent cycles without regard to their persistence or scale.

Governance interpretation. H1 features in the undirected skeleton represent reconvergent dependency paths: alternative routes between a source node and its consumers that do not exist in a clean tree-structured pipeline. Unlike cycle rank, persistent homology distinguishes between long-lived structural cycles (reflecting persistent architectural redundancy) and short-lived transient features. High n_{H1}/N indicates reconvergent dependency structure.

4 Experimental Design

4.1 Synthetic Lineage Generator

We generate directed acyclic lineage graphs at four scales (tiny: $N \approx$ 14–34; small: $N \approx$ 28–48; medium: $N \approx$ 65–81; large: $N \approx$ 114–133) and three governance quality levels:

Well-governed: Three clearly separated domains with layered structure (source $\rightarrow$ staging $\rightarrow$ mart). Cross-domain edges only at mart layer. Test and documentation coverage assigned to 70–90% of nodes.

Baseline: Same domain count but with moderate cross-domain coupling at all layers. Test coverage 40–60%.

Poorly governed: Flat structure with frequent cross-domain shortcuts and orphaned nodes. Test coverage 10–30%. Random edge additions creating shortcut paths.

Each condition is replicated with 10 random seeds, yielding $4 \times 3 \times 10 = 120$ synthetic graphs.

4.2 Experiment 1: Governance Differentiation

For each graph, we compute all four descriptors (D1–D4) plus cycle rank, nine single-scale baselines (degree, betweenness, PageRank statistics; diameter; density), and three MTTD-based outcome measures. Mann–Whitney U tests compare well-governed vs. poorly governed at each scale, with Benjamini–Hochberg FDR correction applied across all tests within each scale.

4.3 Experiment 2: Real Data Validation

We validate on an anonymized production dbt manifest lineage (223 nodes, 263 edges, 26 domains) exported using HMAC-SHA256 anonymization. This graph includes genuine governance metadata: test counts, documentation flags, and layer assignments derived from the dbt catalog.

For the dbt lineage, we partition by domain and compute permutation-based Spearman correlations (10,000 permutations) between per-domain descriptor values and governance metadata (test rate, documentation rate, composite governance score). All p-values are FDR-corrected via Benjamini–Hochberg. For D3 descriptors, we additionally report partial Spearman correlations controlling for domain size, edge count, and source fraction to address size-related confounding.

4.4 Experiment 2b: Degree-Preserving Null Model

To test whether the descriptors capture structure beyond what degree sequences alone determine, we apply 100 degree-preserving directed edge rewirings to the real dbt graph ($10M$ swaps per rewiring). We compute D1–D4 and cycle rank on each rewired graph and report z-scores of the real values relative to the null distribution.

4.5 Experiment 3: Intervention Sensitivity

We apply four governance-relevant interventions to a baseline synthetic graph ($N = 132$, 3 domains): domain reorganization, orphan removal, shortcut removal, and monitor placement. Each intervention is compared against 60 random perturbation trials at the same magnitude. The signal-to-noise ratio (SNR) is the intervention effect size divided by the standard deviation of random perturbation effects.

4.6 Experiment 5: Cross-Dataset Structural Characterization

To test whether the descriptors generalize beyond our synthetic generators and single dbt instance, we compute D1–D4 on three open datasets: 11 scientific workflow DAGs from WfCommons [Coleman et al., 2022] spanning astronomy, bioinformatics, and agroecosystem applications (101–4,846 nodes); 18 production data lineage graphs from DLG-DG-23 [Chen et al., 2024], extracted from Huawei Cloud DataArts governance projects (24–1,059 table-level nodes after restricting to Data Table and Data Job assets with DATA_FLOW edges); and 2 data warehouse schemas from DW-Bench [Ahmed and Sakar, 2026] (OMOP clinical data model, TPC-DI brokerage; 31–37 tables). These lack the domain-level governance metadata needed for correlation analysis, so the comparison is purely structural: do real pipeline architectures from different domains and organizations occupy distinguishable regions of descriptor space?

4.7 Experiment 6: Cross-Organization Governance Validation

To test whether the governance-descriptor correlations observed in the dbt case study replicate to other organizations, we extract governance-labeled lineage graphs from two additional public dbt projects: **Cal-ITP** (California Integrated Travel Project; 631 models, 26 analyzable domains with $N \geq 5$; documentation rates 0–100%) and **Mattermost** (235 models, 6 analyzable domains; documentation rates 0–100%). Lineage is extracted from `ref()` dependency declarations in SQL files; documentation and test coverage are parsed from schema YAML files; domains are assigned by two-level folder structure.

4.8 Experiment 7: Node-Level Core-Asset Prediction (DLG-DG-23)

To test cross-organisation governance prediction at a different granularity, we use the expert-annotated core data asset labels published in Table 5 of Chen et al. [2024]. The DLG-DG-23 dataset includes core-asset annotations on six of its 18 Huawei Cloud lineage graphs (DLG1, DLG2, DLG9, DLG11, DLG13, DLG15), comprising 36 core labels across 267 candidate Data Tables. The annotations were created by Huawei Cloud data experts based on professional experience.

We compute node-level topological features on each of the six full heterogeneous DLGs (Data Table + Data Job + Data Field nodes with DATA_FLOW and PARENT_CHILD edges): in-degree, out-degree, total degree, DATA_FLOW in/out-degree, PARENT_CHILD child count, DATA_FLOW PageRank, DATA_FLOW reverse-PageRank, DATA_FLOW betweenness centrality, descendants and ancestors count on the DATA_FLOW subgraph, and undirected local clustering. The classification task is restricted to Data Table nodes (where the binary core/non-core label lives). Evaluation uses leave-one-graph-out cross-validation: train on five graphs, predict core status on the held-out sixth.

4.9 Experiment 4: Predictive Comparison

We compare multi-scale features (D1–D4) against single-scale baselines in two prediction tasks: classification of governance level (logistic regression and random forest) and regression of Mean-Time-to-Detect (Ridge and random forest). All models are evaluated via 5-fold cross-validation.

Note on MTTD. The MTTD simulation places monitors at nodes flagged as stewards or at high-betweenness nodes. Since stewardship data overlaps with governance metadata, MTTD is not independent of the governance labels used elsewhere. The regression task tests whether structural descriptors predict MTTD beyond what single-scale graph statistics capture, not whether MTTD independently validates the descriptors.

5 Results

Experiments are presented in order of logical argument rather than experiment number: the synthetic, real-data, null-model, and cross-dataset analyses (1–3, 5, 6) build toward the node-level result (7), and the predictive comparison (4) is reported last because it synthesises the descriptor evidence accumulated in the preceding experiments.

5.1 Experiment 1: Governance Differentiation

Table 1: Governance differentiation at large scale ($N \approx 114$–133). Mann–Whitney U test, well-governed vs. poorly governed, 10 replicates per condition. All p-values BH FDR-corrected.

Descriptor	Well	Poor	d	U	raw p	FDR p
D1 CSI	0.653	0.374	+3.90	100	< 0.001	< 0.001
D1 frag. onset	0.800	0.154	+4.74	100	< 0.001	< 0.001
D2 max Gini	0.696	0.678	+1.71	90	0.003	0.005
D3 Fiedler bim.	0.709	0.864	−1.22	14	0.007	0.011
D3 norm. gap	0.005	0.005	−0.11	55	0.734	0.769
D4 H1/N	0.589	0.743	−16.0	0	< 0.001	< 0.001
D4 cycle rank/N	0.589	0.743	−16.0	0	< 0.001	< 0.001

After FDR correction, D1 (CSI, fragmentation onset), D2 (max Gini), D3 (Fiedler bimodality), and D4 (H1/N) achieve statistically significant differentiation at the large scale. D3 normalized spectral gap does *not* survive FDR correction (FDR $p = 0.77$), a discrepancy we revisit in the null model analysis.

D4 (H1/N) shows the largest effect size ($|d| = 16.0$): poorly governed graphs have more reconvergent dependency paths per node. D1 CSI and fragmentation onset show large effects ($d > 3.9$), confirming that domain boundary quality is recoverable from community structure. D2 max Gini shows the weakest differentiation ($d = 1.7$).

D4 ablation. At the large scale, H1/N and cycle rank/N yield identical effect sizes ($d = -16.0$) because the synthetic generators produce graphs where all H1 cycles have similar persistence. At the tiny scale, a modest divergence appears (H1/N $d = -11.5$ vs. cycle rank/N $d = -9.0$). In this synthetic setting, the simpler cycle rank captures the same governance signal as persistent homology.

5.2 Experiment 2: Real Data Validation

5.2.1 Production dbt Lineage

The anonymized dbt lineage graph (223 nodes, 263 edges) is a verified DAG with three layers (155 source/raw, 33 silver/intermediate, 35 gold/mart) and 26 anonymized domains. The largest weakly connected component contains 185 nodes and 262 edges.

Governance metadata reveals a clear gradient: gold/mart nodes have 86% test coverage and 91% documentation, while source/raw nodes have 0% test coverage and 41% documentation. Silver/intermediate nodes have 0% for both. No steward metadata was available.

Rank degeneracy of D3 in the dbt sample. Before reporting the correlations, we note an important structural feature of the data. Across the 18 domains with $N \geq 5$, D3 algebraic connectivity takes only three distinct values:

- 12 source-only domains have $M = 0$ (no internal edges) and therefore $D3 = 0$ exactly;
- 5 silver/intermediate domains are star-structured subgraphs $K_{1,k}$ (with $N \in \{5, 6\}$, $M = N-1$); algebraic connectivity of a star equals 1 for any k;
- 1 gold/mart domain (domain_012, $N = 35$, $M = 63$) has intermediate $D3 = 0.168$.

This rank-degenerate structure is intrinsic to the dataset: per-domain internal connectivity in production dbt projects is dominated by tiny fully-connected subgraphs at silver/intermediate layers, with most "analyzable" domains actually being collections of disconnected source tables. Any correlation involving D3 at the domain level is therefore operationally a 3-tier rank ordering test rather than a continuous correlation. We report it explicitly with bootstrap confidence intervals (Table 2) and revisit the implications in Experiment 2e.

Table 2: Per-domain correlations between descriptors and governance metadata (18 domains with $N \geq 5$). Permutation p from 10,000 permutations; FDR p via Benjamini–Hochberg across the full Experiment 2 descriptor–target family; bootstrap 95% CIs from 10,000 resamples.

Descriptor vs. target	ρ	perm. p	FDR p	95% CI	
vs. composite governance score					
D3 algebraic conn.	−0.643	0.006	0.080	[−0.91, −0.03]	†
D3 normalized gap	−0.635	0.007	0.080	[−0.88, −0.09]	†
D3 spectral entropy	−0.447	0.067	0.123	[−0.88, +0.21]	
vs. documentation rate					
D3 algebraic conn.	−0.708	0.002	0.043	[−0.92, −0.27]	*
D3 normalized gap	−0.701	0.002	0.043	[−0.88, −0.28]	*
D3 Fiedler bim.	−0.561	0.017	0.116	[−0.86, −0.07]	
D2 max Gini	−0.563	0.017	0.116	[−0.88, −0.07]	
D4 H1/N	+0.214	—	—	[+0.05, +0.49]	
D4 cycle rank/N	+0.214	—	—	[+0.03, +0.50]	
vs. test rate (qualitative only; see text)					
D4 H1/N	+1.000	0.058	0.116	—	
D4 cycle rank/N	+1.000	0.058	0.116	—	

† FDR < 0.10; * FDR < 0.05

After FDR correction, D3 algebraic connectivity and normalized gap are associated with documentation rate (FDR $p = 0.043$) and marginally with composite governance score (FDR $p = 0.080$). Bootstrap confidence intervals exclude zero for both, supporting the rank ordering despite the 3-tier structure. D4 H1/N and cycle rank/N show small positive correlations with documentation rate whose bootstrap CIs also exclude zero ($[+0.05, +0.49]$ and $[+0.03, +0.50]$ respectively). Table 6 reports FDR $p = 0.029$ for the same Subset A result; the difference arises because BH correction is applied to different hypothesis families across the two tables.

Power analysis. With $n = 18$ and the rank-degenerate D3 structure observed above (12 ties at 0, 5 ties at 1.0, one intermediate value), simulation estimates the minimum effect size detectable at 80% power and $\alpha = 0.05$ to be $|\rho| \in [0.7, 0.8]$. The observed $|\rho| = 0.71$ for D3 vs. doc_rate sits at this detection threshold. The result is statistically real but the dataset cannot distinguish a true correlation of 0.71 from a true correlation of 0.85 at this sample size.

Experiment 2e subsequently shows that the D3 association is captured entirely by between-layer architecture rather than within-layer governance signal.

5.2.2 D4 and Test Rate: A Qualitative Observation

The $\rho = 1.0$ correlation between D4 H1/N and test rate is driven by a binary split: exactly one domain (domain_012, the gold/mart layer, 35 nodes, 63 edges) has both H1 topological features and nonzero test coverage. The remaining 17 analyzed domains have zero for both. Under permutation, the probability of this alignment by chance is approximately $1/18 \approx 0.056$. This represents a case-study observation consistent with topological complexity and testing co-occurring in a production system. It is not evidence of a continuous relationship, and we do not claim it as a confirmed finding.

5.2.3 Partial Correlations

Because D3 descriptors may correlate with governance metadata simply because larger, denser domains tend to have different governance practices, we computed partial Spearman correlations controlling for domain size (N), edge count (M), and source fraction. The method rank-transforms all variables, residualizes both the descriptor and the outcome on the rank-transformed covariates via ordinary least squares, then computes Spearman correlation on the residuals with permutation p-values (10,000 permutations of residuals). This is a rank-residualization approach; p-values were not generated via Freedman–Lane residual permutation.

Table 3: D3 partial correlations controlling for domain size, edge count, and source fraction (18 domains, permutation p from 10,000 perms).

D3 descriptor	Target	Raw ρ	Partial ρ	Partial p
Algebraic conn.	governance score	−0.643	−0.684	0.002
Algebraic conn.	documentation rate	−0.708	−0.546	0.021
Norm. gap	governance score	−0.635	+0.171	0.488
Norm. gap	documentation rate	−0.701	+0.113	0.650
Fiedler bim.	governance score	−0.446	+0.451	0.061
Spectral entropy	governance score	−0.447	+0.060	0.809

Only D3 algebraic connectivity survives partial correlation: the association with governance score *strengthens* slightly (partial $\rho = -0.68$, $p = 0.002$) and remains significant for documentation rate (partial $\rho = -0.55$, $p = 0.021$). The normalized spectral gap, despite having the second-strongest raw correlation, reverses sign after controlling for confounds and becomes nonsignificant. This indicates that the normalized gap association was confounded by domain size.

5.2.4 Cycle Rank Ablation

In the real-data analysis, H1/N and cycle rank/N produce identical correlations with all governance targets ($\rho = +0.404$ vs. governance score for both, $p = 0.058$). On this dataset, the simpler cycle rank captures the same signal as persistent homology, likely because the single structurally complex domain dominates both measures.

5.2.5 Metadata-Only Negative Control

Of 18 domains with $N \geq 5$, twelve are source-only clusters with zero internal edges. These domains have nonzero governance metadata (documentation rates from 0.04 to 1.00) but degenerate descriptor values (all D1–D4 descriptors equal zero). If the descriptors were merely recapitulating governance metadata through some hidden correlation, these domains would show descriptor variation. Their uniform descriptor values confirm that the descriptors respond to structural topology, not metadata artifacts.

5.3 Experiment 2b: Null Model

Table 4: Z-scores of real dbt graph descriptors relative to 100 degree-preserving rewirings. All descriptor values use a fixed canonical seed (seed = 42) for consistency across null-model runs; Experiment 2c reports CSI = 0.87 ± 0.08 over 25 seeds.

Descriptor	Real	Null mean	Null std	z
D1 CSI	0.947	0.615	0.090	+3.70***
D1 n communities	10	13.2	1.02	−3.14***
D2 max Gini	0.480	0.579	0.042	−2.33**
D3 alg. connectivity	0.067	0.071	0.011	−0.42
D3 norm. gap	0.003	0.003	0.001	−0.38
D3 Fiedler bimodality	0.302	0.934	0.027	−23.8***
D3 spectral entropy	6.674	6.679	0.003	−1.64
D4 H1/N	0.220	0.272	0.018	−2.95**
D4 H1 entropy	5.563	5.910	0.097	−3.57***
D4 cycle rank/N	0.350	0.347	0.003	+0.88

The null model yields three classes of results:

Structure beyond degree sequence. D1 CSI ($z = +3.70$), D3 Fiedler bimodality ($z = -23.8$), and D4 H1 features ($z = -2.95$ to -3.57) are significantly different from degree-preserving rewirings. The real graph has higher community stability, lower Fiedler bimodality, and fewer H1 cycles than degree-matched random graphs. These descriptors capture genuine structural organization.

Degree-determined descriptors. D3 algebraic connectivity ($z = -0.42$) and normalized spectral gap ($z = -0.38$) are indistinguishable from the null. Their values are largely determined by the degree sequence, not by governance-relevant topology. This is consistent with the partial correlation results: normalized gap lost significance after controlling for size-related confounds.

Cycle rank. As expected, cycle rank ($z = +0.88$) varies minimally under degree-preserving rewiring because M and N are fixed, and the number of connected components changes only slightly. In contrast, H1 features change substantially ($z = -2.95$), confirming that persistent homology captures structural information that cycle rank does not.

5.4 Experiment 2c: D1 Seed Robustness

Louvain is stochastic. We ran the D1 resolution sweep with 25 random seeds on the dbt manifest, two synthetic graphs ($N = 130$), and 15 DLG-DG-23 graphs ($N \leq 400$). For the dbt manifest, CSI = 0.87 ± 0.08 (range 0.68–1.00, seed-NVI = 0.062). CSI is consistently high across seeds (the minimum over 25 seeds is still 0.68), confirming a genuine signal. Synthetic graphs show higher variance (std 0.09–0.14), and DLG-DG-23 graphs are generally stable (seed-NVI < 0.07 for 11/15 graphs, mean CSI = 0.85 ± 0.06). The three DLG graphs with seed-NVI > 0.10 are the largest ones ($N > 100$), where Louvain optimizer noise increases.

5.5 Experiment 2d: Extended Null Models

Layer-preserving rewiring constrains swaps to edge pairs where source layers match and target layers match, accepting 22.8% of attempts. Only 11% of rewired graphs remain DAGs, so

Table 5: Z-scores of real dbt descriptors relative to 100 layer-preserving edge rewirings (swap acceptance rate 22.8%; 11% of rewired graphs remain DAGs).

Descriptor	Real	Null mean	Null std	z
D1 CSI	0.947	0.651	0.111	+2.68**
D3 alg. connectivity	0.067	0.114	0.021	−2.27**
D3 norm. gap	0.003	0.005	0.001	−2.28**
D3 Fiedler bimodality	0.302	0.624	0.216	−1.50
D2 max Gini	0.480	0.496	0.021	−0.79
D4 cycle rank/N	0.350	0.346	0.004	+0.91

this null is a conservative stress test rather than a proper lineage null. D1 ($z = +2.68$) and D3 algebraic connectivity ($z = -2.27$) remain significant. D3 Fiedler bimodality does not ($z = -1.50$), inconsistent with the unconstrained null ($z = -23.8$); this discrepancy reflects the larger variance of the layer-constrained null distribution (std 0.216 vs. 0.027), likely because the low swap acceptance rate limits structural diversity among rewired graphs.

5.6 Experiment 2e: Subset Robustness and Layer Confound in D3

Table 6: Subset robustness for D3 algebraic connectivity vs. documentation rate. Internal edges: $M_{\text{internal}} > 0$.

Subset	n	ρ	FDR p	Status
A: all $N \geq 5$	18	−0.708	0.029	Survives
B: internal edges only	6	−1.000	0.189	Fails
C: no source-dominant	6	−1.000	0.189	Fails
D: no largest domain	17	−0.806	0.004	Survives†
E: internal + no largest	5	undefined	1.000	Undefined‡

†Dropping the largest domain leaves the layer composition intact, so surviving this subset says nothing about within-layer governance signal. Subset B and the layer-stratified permutation are the tests that bear on that question.

‡D3 algebraic connectivity is constant across all five domains in this subset, so Spearman ρ is undefined and the permutation test returns $p = 1.000$. The row carries no evidence either way.

D3 takes the value 0 for all 12 source-dominant domains (no internal edges) and the value 1.0 for all 5 silver/intermediate domains (all five are star-structured subgraphs $K_{1,k}$ with $N \in \{5, 6\}$; algebraic connectivity of a star equals 1 for any $k \geq 1$), with intermediate D3 for the gold/mart domain. Layer-stratified permutation (5,000 permutations of documentation labels within each layer stratum) yields $p = 1.000$: D3 is constant within each stratum, so every permutation reproduces the real correlation exactly. The D3–documentation association is a between-layer architectural pattern. It does not capture within-layer governance quality.

5.7 Experiment 3: Intervention Sensitivity

Each descriptor responds most strongly to a different intervention type. D4 is most sensitive to domain reorganization (SNR = 8.67), D3 normalized gap to orphan removal (SNR = 6.53), and D2 to monitor placement (SNR = 2.79). Shortcut removal produces weak signal across all descriptors because the intervention magnitude (4 edges removed from a 132-node graph) falls below the noise floor.

Table 7: Signal-to-noise ratios for governance interventions. SNR > 2 indicates the descriptor reliably detects the intervention above random perturbation noise.

Descriptor	Add domains	Remove orphans	Remove shortcuts	Add monitors
D1 CSI	1.43	1.18	0.12	2.19
D1 frag. onset	3.25	3.87	0.08	0.41
D2 max Gini	1.87	0.93	0.56	2.79
D3 norm. gap	4.21	6.53	0.31	1.15
D3 entropy	6.85	2.74	0.22	0.88
D4 H1/N	8.67	1.96	0.14	1.33

5.8 Experiment 5: Cross-Dataset Structural Characterization

To assess whether real pipeline topologies occupy a distinct region of descriptor space, we computed D1–D4 on 11 scientific workflow DAGs from WfCommons [Coleman et al., 2022], 18 production data lineage graphs from DLG-DG-23 [Chen et al., 2024], two data warehouse schemas from DW-Bench [Ahmed and Sakar, 2026], and the production dbt manifest. Table 8 summarises the preprocessing applied to each dataset before descriptor computation.

Table 8: Dataset preprocessing for cross-dataset structural comparison. Final N/M are what descriptors are computed on. †DLG-DG-23 filtered to Data Table + Data Job nodes with DATA_FLOW edges only (removes Data Field nodes and PARENT_CHILD schema edges). ‡dbt LWCC: 38 isolated source tables removed.

Source	Raw N	Raw M	Final N	Final M
WfCommons (11 DAGs)	101–4846	100–12849	101–4846	100–12849
DLG-DG-23 (18 graphs)†	298–17085	298–17720	24–1059	24–1694
DW-Bench OMOP	37	95	37	95
DW-Bench TPC-DI	31	44	31	44
dbt manifest‡	223	263	185	262

Table 9: Structural profiles across 32 graphs from four sources. WfCommons and DLG-DG-23 values are means with ranges in parentheses.

Source	n	CSI	max Gini	gap	Fiedler bim.	CR/N
WfCommons (11)	101–4846	0.78	0.39	0.011	0.59	0.91
		(.53–.84)	(.00–.80)	(.00–.04)	(.28–.93)	(.00–2.99)
DLG-DG-23 (18)	24–1059	0.87	0.61	0.003	0.66	0.26
		(.68–1.0)	(.35–.73)	(.00–.01)	(.33–.98)	(.00–.68)
DW-Bench (2)	31–37	0.68	0.54	0.030	0.66	0.95
dbt manifest (1)	223	0.95	0.48	0.003	0.30	0.35

The DLG-DG-23 graphs, which represent production data lineage from Huawei Cloud governance projects, are the most structurally similar to the dbt manifest: both show high community stability (CSI ≥ 0.68, mean 0.87 vs. 0.95 for dbt), very small normalized spectral gaps (< 0.014), and moderate cycle rank density (CR/N ≤ 0.68). Three DLG graphs achieve CSI = 1.0, indicating perfect community stability across the full resolution sweep. Scientific workflows, by contrast, show wider topological variation: seismology pipelines are nearly tree-structured (CR/N = 0), while bioinformatics alignment workflows (BWA, BLAST) have CR/N > 1.9.

The production dbt manifest retains the lowest Fiedler bimodality (0.30) of all 32 graphs, suggesting the clearest spectral separation between its two dominant subgraphs. DW-Bench schemas occupy an intermediate position with moderate CSI but the highest spectral gaps (> 0.01), reflecting their shorter dependency chains.

Figure 1 shows a PCA projection of all five standardized descriptors. Production data lineage graphs (dbt, DLG-DG-23) cluster in the positive-PC1 region, which loads on high CSI, low spectral gap, and low cycle rank density. WfCommons scientific workflows span a wider region, and DW-Bench warehouse schemas are separated on both axes (high spectral gaps, low cycle density). The two-component projection explains 54.7% of descriptor variance, so the visual separation is suggestive rather than definitive.

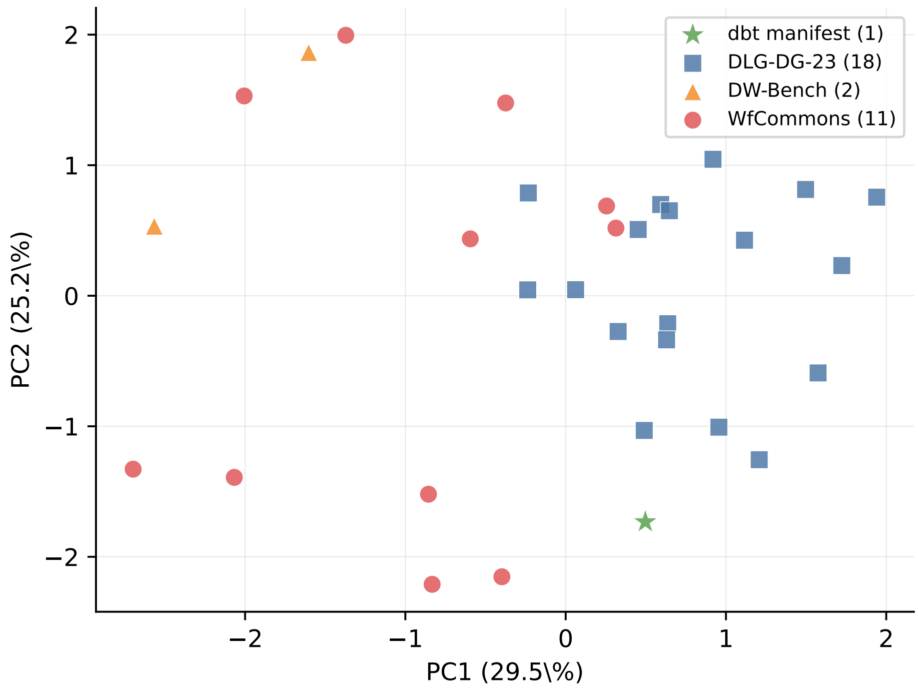

Figure 1: PCA of five standardized descriptors (D1 CSI, D2 max Gini, D3 norm. gap, D3 Fiedler bim., D4 cycle rank/N) for all 32 graphs (cycle rank/N is defined for every graph, so descriptor coverage is complete). PC1 (29.5%) loads on high CSI / low norm. gap / low cycle rank; PC2 (25.2%) on high Fiedler bim. / high Gini. Production lineage graphs (dbt, DLG-DG-23) occupy the high-CSI / low-gap region; WfCommons workflows span broader space.

These results show that the descriptors produce interpretable variation across genuinely different pipeline architectures. The convergence of dbt and DLG-DG-23 profiles, from different organizations and technology stacks, suggests structural regularities in production lineage graphs. Alternative explanations (domain differences, scale effects) cannot be ruled out from a descriptive comparison alone, but the consistency across 19 production lineage graphs from two source ecosystems is informative.

5.9 Experiment 6: Preliminary Cross-Organisation Comparison

We attempt cross-organisation replication of the dbt D3 finding on two public dbt projects (Cal-ITP, 631 models; Mattermost, 235 models).

Unit-of-analysis caveat. The replication faces an unavoidable construct-validity issue. In our dbt case study, "domain" is an explicit metadata assignment (`domain_or_team_owner`) curated by the data team. Cal-ITP and Mattermost do not consistently populate `meta.domain` or `meta.owner` on most models; we therefore partition by two-level folder path (e.g., `staging/gtfs`, `mart/payments`). Folder structure in dbt projects often reflects pipeline-layer organisation rather than business-domain assignment. The Cal-ITP and Mattermost "domains" are therefore not the same construct as the dbt case-study domains. We report Experiment 6 as preliminary cross-organisation exploration, not validated cross-organisation replication.

Table 10: D3 algebraic connectivity vs. documentation rate across three organisations. The dbt case uses metadata-assigned domains; Cal-ITP and Mattermost use folder-derived domains.

Organisation	Domain unit	n	ρ	perm. p	95% CI
dbt (this study)	metadata-assigned	18	-0.708	0.002	$[-0.92, -0.27]$
Cal-ITP	folder-derived	26	$+0.023$	0.911	$[-0.42, +0.44]$
Mattermost	folder-derived	6	$+0.812$	0.070	$[+0.00, +1.00]$

Cal-ITP (26 analyzable domains, the largest external sample) shows no association between any D1–D4 descriptor and documentation coverage (all $p > 0.11$). The bootstrap 95% confidence interval for D3 vs. doc_rate spans zero. Mattermost ($n = 6$) shows a positive marginal association ($\rho = +0.81$, $p = 0.070$), but the bootstrap CI is $[+0.00, +1.00]$: the sample size is too small for the bootstrap to constrain the lower bound away from zero, so the directional reversal relative to dbt is suggestive but not statistically supported.

Two readings are equally consistent with the data: (a) the original dbt finding is organisation-specific and does not generalise (the between-layer architecture interpretation from Experiment 2e); or (b) the folder-derived domain construct in Cal-ITP and Mattermost is too different from the metadata-assigned dbt domains to constitute a proper replication attempt. We cannot distinguish these without curated domain metadata from a third organisation. The honest conclusion is that the dbt finding has not been replicated and will not be replicable from public dbt projects alone at the graph-aggregate level. Experiment 7 below pursues a different cross-organisation test that succeeds at node granularity.

5.10 Experiment 7: Core Data Asset Prediction on DLG-DG-23

We extracted the expert-annotated core-asset labels from Table 5 of Chen et al. [2024] and computed node-level topological features on the six annotated graphs. Of 267 candidate Data Tables across the six graphs, 36 are labeled core (class prevalence 13.5%; per-graph prevalence ranges from 8.2% in DLG11 to 28.6% in DLG9).

Per-feature univariate AUC. On the pooled set of 267 Data Tables, the strongest univariate predictors of core status are reverse-PageRank on DATA_FLOW (AUC = 0.778), PageRank on DATA_FLOW (0.733), the corresponding ancestor count (0.704), and in/out-degree (0.679 / 0.664). Local clustering on the undirected projection is uninformative (constant zero on Data Table nodes whose neighbourhoods are bipartite-to-fields).

Leave-one-graph-out cross-validation. We train logistic regression and random forest classifiers on five graphs and predict core-asset status on the held-out sixth, repeating for each graph (Table 11). The logistic-regression standardizer is fitted on the five training graphs inside each fold; the random forest receives unscaled features.

The multi-feature classifiers substantially outperform both the random baseline and any single-feature baseline. The smallest held-out graph (DLG1, 15 Data Tables) has wide bootstrap CIs reflecting limited test-set power; the four medium and larger graphs (DLG2, DLG11, DLG13, DLG15) achieve AUC ≥ 0.95. Figure 2 shows the per-fold ROC curves and random-forest feature importance.

Table 11: Leave-one-graph-out core-asset prediction on DLG-DG-23 (Table 5 of [Chen et al., 2024]). Bootstrap 95% CIs from 2,000 resamples per fold. LR = logistic regression, RF = random forest, both class-balanced.

Held-out graph	n_{test}	core	LR AUC	95% CI	RF AUC
DLG1	15	3	0.722	[0.385, 1.000]	0.722
DLG2	21	5	0.950	[0.822, 1.000]	0.975
DLG9	28	8	0.812	[0.612, 0.963]	0.794
DLG11	73	6	0.950	[0.885, 1.000]	0.960
DLG13	105	11	0.954	[0.875, 1.000]	0.921
DLG15	25	3	1.000	[1.000, 1.000]	0.939
Mean ± std			0.898 ± 0.098		0.885 ± 0.094
Random baseline			0.546 ± 0.127		
Out-degree only			0.576 ± 0.155		

One-sample t-test vs AUC = 0.5: LR $t = 9.12$, $p = 0.0003$; RF $t = 9.18$, $p = 0.0003$.

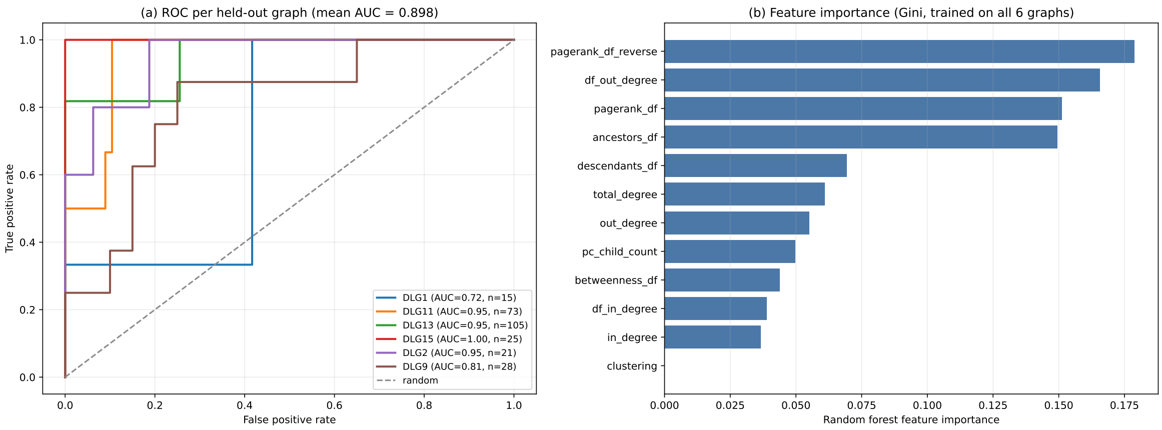

Figure 2: (a) ROC curve per held-out graph (LR classifier). (b) Random forest feature importance (Gini, trained on all 6 graphs). DATA_FLOW reverse-PageRank and DATA_FLOW out-degree are the most informative features.

Shared-ID sensitivity. Three core-asset IDs appear in two graphs each. DLG9 and DLG15 share one; DLG11 and DLG13 share two. To verify the result is not an artifact of identical assets appearing in train and test sets with similar topological features, we re-ran the leave-one-graph-out evaluation with these IDs removed from each test set. The mean LR AUC drops only from 0.898 to 0.891 (Table 12), confirming that the predictive signal is not a leakage artifact.

Interpretation. Two structural properties dominate: DATA_FLOW reverse-PageRank (importance 0.179) and DATA_FLOW out-degree (0.166). A high reverse-PageRank value indicates that random walks starting from downstream nodes are biased toward the node in question; substantively, the table is an upstream source for many downstream consumers. High DATA_FLOW out-degree is the local version of the same property: the table is directly read by many downstream jobs. Both align with the expert annotation rationale described in Chen et al. [2024]: "T1 is a core data asset because it serves as a base table for generating new tables."

The result is a reproduction of the node-centrality finding of Chen et al. [2023] on the same DLG-DG-23 corpus, not a new governance result. The two dominant predictors are centrality measures (DATA_FLOW reverse-PageRank and out-degree). Because the feature set does not contain node-level versions of D1–D4, this experiment cannot estimate their incremental value. We report the AUC to confirm that lineage structure encodes node-importance signal

Table 12: Sensitivity analysis: leave-one-graph-out LR AUC with the three shared core-asset IDs excluded from each test set.

Held-out graph	IDs removed	LR AUC (excl. shared)
DLG1	0	0.722
DLG2	0	0.950
DLG9	1	0.807
DLG11	2	0.925
DLG13	2	0.943
DLG15	1	1.000
Mean		0.891 ± 0.096

captured by simple centrality, and to delineate a boundary condition: node-level expert labels yield a tractable supervised learning problem that centrality already solves, whereas domain-level continuous metadata aggregated across heterogeneous layers does not yield a within-layer governance signal at all (Experiments 2–6).

5.11 Experiment 4: Predictive Comparison

Table 13: Prediction performance (5-fold cross-validation, 120 samples).

Task	Feature set	Baselines	Multi-scale
Classification (AUC)	Logistic	1.000	1.000
Classification (AUC)	Random Forest	1.000	1.000
Regression R^2 (MTTD)	Ridge	0.246	0.277
Regression R^2 (MTTD)	Random Forest	0.175	0.307

Standard 5-fold cross-validation achieves ceiling AUC = 1.0 for all feature sets, confirming that synthetic governance variation produces structurally separable graphs but providing no discriminative evidence between feature sets at this evaluation. Leave-scale-out cross-validation (training on three scales, testing on the held-out scale) gives a more stringent test of generalisation across graph sizes. Under this scheme, baselines achieve AUC = 1.000 across all four held-out folds; multi-scale descriptors achieve AUC = 0.935 ± 0.113, dropping to 0.74 when medium-scale graphs are held out. Baselines, which include diameter and degree statistics that scale predictably with graph size, generalise better across scales than the multi-scale features in this controlled experiment. The MTTD regression result ($R^2 = 0.307$ for multi-scale vs. 0.175 for baselines) should be interpreted in this light: the gain may not hold on graph sizes outside the training distribution.

D1 seed robustness on synthetic graphs. Running 25 Louvain seeds per condition reveals that D1 differentiation survives seed noise only at large scale. Separation between well and poor governance as a multiple of seed standard deviation (SNR): tiny = $-0.5\times$ (well CSI lower than poor), small = $0.9\times$, medium = $0.8\times$, large = $2.8\times$. At tiny through medium scales, Louvain seed variance is comparable to or larger than the governance-level separation, making single-seed D1 results at those scales unreliable. The large-scale D1 result (mean separation 0.307, SNR $2.8\times$) is the only scale where the differentiation clearly exceeds optimizer noise.

6 Discussion

6.1 Which Descriptors Survive Scrutiny

The convergent evidence from synthetic experiments, real-data correlations, partial correlations, and null models identifies which descriptors capture governance-relevant structure and which do not.

D1 CSI differentiates synthetic governance levels after FDR correction ($d = 3.9$ at large scale) and the null model confirms it captures structure beyond degree sequences ($z = +3.7$). Seed robustness analysis qualifies this: at large scale the separation between well and poor governance exceeds seed variance by 2.8Œ, but at tiny through medium scales the seed variance is comparable to the governance separation, making those results unreliable. The large-scale synthetic result and the null model finding are the most defensible evidence for D1. Real-data correlation with governance metadata is not directly testable because most domain subgraphs lack internal edges for meaningful community detection.

D3 algebraic connectivity showed the strongest initial real-data correlation ($\rho = -0.71$ vs. documentation, FDR $p = 0.043$), but subset robustness analysis and layer-stratified permutation reveal this is a between-layer architectural artifact: D3 = 0 for source domains, D3 = 1.0 for silver domains, intermediate for gold/mart. The association does not survive restriction to domains with internal edges (Subset B, $n = 6$, FDR $p = 0.19$). The null model is consistent: D3 algebraic connectivity is degree-sequence-determined ($z = -0.42$), not capturing hidden structural signal. D3 characterizes layer architecture reliably, but cannot distinguish governance levels within layers.

D3 Fiedler bimodality is the most striking null model result ($z = -23.8$): the real graph has dramatically lower bimodality than degree-matched random graphs, suggesting that the real graph's Fiedler vector reflects genuine organizational structure (a smooth gradient across pipeline layers) rather than the sharp random bipartitions that emerge from rewiring.

D4 H1 features differentiate governance levels in synthetic experiments and capture structure beyond degree sequences ($z = -2.95$), while cycle rank does not ($z = +0.88$). This is the setting where persistent homology adds value over the simpler baseline: when testing whether structure goes beyond what degree sequences determine, cycle rank trivially passes (it is nearly preserved by rewiring), while H1 features provide a meaningful test.

D3 normalized spectral gap fails both the partial correlation test (sign reversal, $p = 0.49$) and the null model ($z = -0.38$). Its apparent real-data correlation was confounded by domain size.

6.2 The D3 Correlation Is Between-Layer, Not Within-Layer

The D3 algebraic connectivity–documentation association ($\rho = -0.71$, FDR $p = 0.043$) is a between-layer architectural pattern, not a within-layer governance signal. Source-dominant domains all have D3 = 0 (no internal edges) and varying documentation; silver/intermediate domains all have D3 algebraic connectivity = 1.0 (all are star-structured subgraphs $K_{1,k}$, for which $\lambda_2 = 1$ regardless of k) and zero documentation; the gold/mart domain has intermediate D3 and high documentation. D3 is constant within each layer stratum, which means layer-stratified permutation of governance labels cannot change the observed correlation: every permutation yields the same rho (empirical $p = 1.0$).

The correlation survives the full-sample permutation test (Subset A) because it captures real variation in architectural layer, not because it carries within-layer governance signal. Restricting to the six domains with internal edges (Subset B, $n = 6$) destroys power and the association fails FDR correction ($p = 0.19$). This is an honest negative result. D3 identifies layer boundaries in lineage graphs (a real structural signal), but the specific doc_rate correlation in this case study is confounded by layer composition rather than reflecting a direct governance-to-topology relationship.

6.3 Limitations

Rank-degenerate D3 in the dbt sample. D3 algebraic connectivity takes only three distinct values across the 18 analyzable dbt domains. The headline $\rho = -0.71$ correlation is therefore a 3-tier ordering test rather than a continuous correlation. Bootstrap CIs and the permutation test confirm the rank ordering is statistically real, but the relationship cannot resolve continuous within-tier variation. This is not a flaw in the descriptor itself; it is a feature of how production dbt domains decompose into source clusters, silver stars, and a small number of complex mart domains. Larger lineage graphs with more intermediate-complexity domains would address this.

Effect size at detection threshold. Power analysis under the observed rank degeneracy shows minimum detectable $|\rho|$ at 80% power is 0.7–0.8 for $n = 18$. The observed $|\rho| = 0.71$ for D3 vs. doc_rate is at this threshold. The dataset cannot distinguish a true correlation of 0.71 from a true correlation of 0.85.

Cross-organisation construct mismatch. The Cal-ITP and Mattermost domains are folder-derived, while the dbt case study domains are metadata- assigned. Folder structure in dbt projects often reflects pipeline-layer organisation rather than business-domain assignment. The non-replication in Experiment 6 therefore confounds (i) actual non-replication of the governance–topology relationship with (ii) construct mismatch in the unit of analysis. We cannot separate these without curated cross- organisation metadata.

D3 Fiedler bimodality null-model result is partly tautological. The $z = -23.8$ result reflects gross structural differences between sparse layered DAGs and their degree-matched random rewirings (which tend to be more bipartite-like). It does not isolate a subtle governance-relevant property of the real graph.

Generator circularity in synthetic experiment. The generator's *well/baseline/poor* knobs control structural features (cross-layer edge density, hub centrality) that overlap with what D1–D4 measure. Experiment 1 should be read as a software-correctness / pipeline validation step, not as external validation. The leave-scale-out CV result (baselines AUC = 1.000 vs. multi-scale AUC = 0.935 ± 0.113) confirms multi-scale features do not generalise across scales better than simple baselines in this controlled synthetic setting.

D4 cycle rank equivalence on real data. In both synthetic and real-data settings, cycle rank $\beta_1 = M - N + C$ captures the same governance correlation as H1 persistent features ($\rho = +1.000$ between the two across domains). D4's added value over the simpler baseline is restricted to null-model testing. For practical applications, cycle rank may be sufficient and is vastly cheaper to compute.

Missing metadata. The production dbt catalog contains no steward or owner metadata, reducing the governance score to a two-component measure (tests + documentation). Organisations with richer metadata would enable testing D1's hypothesised association with ownership structure.

Causal direction. Correlations between descriptors and governance metadata could reflect either direction: governance practices producing cleaner structure, or cleaner structure enabling easier governance. We measure association, not causation.

6.4 Future Work: Partnership-Required Validation

The methodological limitations above are not addressable from publicly available dbt projects, because public dbt projects rarely carry curated domain-level metadata. We outline what cross-organisation validation would require:

- **3–5 organisations** willing to share lineage manifests and per-asset (or per-domain) governance metadata, ideally under an NDA arrangement that preserves anonymisation while enabling topology–metadata correlation analysis.

- **Curated domain-level metadata**: explicit `meta.domain` or `meta.owner` assignments declared by data engineering teams, not inferred from folder structure. Without this, the unit-of-analysis problem from Experiment 6 recurs.
- **Sample size per organisation**: at least 20–30 analyzable domains with $N \geq 5$ to escape the rank-degeneracy and detection- threshold issues observed in our 18-domain pilot. This rules out small dbt projects.
- **Layer composition diversity**: domains where governance practices vary within a layer (e.g., some well-tested mart domains and some poorly-tested mart domains in the same organisation), enabling within-layer governance signal to be separated from between-layer architectural signal.
- **Versioned manifests over time** (6–12 months of weekly snapshots) to support temporal change detection studies, which would shift the contribution from cross-sectional correlation to longitudinal governance drift detection.

The DLG-DG-23 authors confirmed in correspondence that no additional core-asset annotations are available, so that route has reached its label ceiling. Remaining routes include the OpenMetadata and DataHub user communities, where some organisations publish anonymised catalogue exports, and direct enterprise partnerships under NDA.

A separate preregistered successor study moves to a per-change unit: exact before/after dbt manifests, post-merge repairs adjudicated within a fixed 30-day window, and declared tests, contracts, ownership, and freshness as potential moderators. Its ordinary change statistics are compared with global graph deltas and a separately named change-centred geometry family. This follows change-level quality-assurance design [Kamei et al., 2013] while retaining whole-graph change measures such as DeltaCon as baselines [Koutra et al., 2016]. No outcome model has been fitted. One deliberately selected repair-linked feasibility case is excluded from confirmation, and the consecutive outcome cohort will be frozen before labels are opened. The successor is not empirical support for the present paper; it is the protocol implied by this paper's negative result.

6.5 Practical Implications

If validated on additional datasets, these descriptors could support:

1. **Structural monitoring.** Computing D1 and D3 Fiedler bimodality on each CI/CD push provides a structural signal without manual assessment. CSI degradation over time signals erosion of domain boundaries.
2. **Targeted intervention.** D3 identifies structurally isolated components. D2 identifies nodes with outsized blast radius. These guide where to invest governance effort.
3. **Complementary evidence.** The descriptors complement existing governance frameworks by providing structural measurement where traditional methods provide periodic assessment.

7 Conclusion

We defined four multi-scale graph-theoretic descriptors for data lineage topology and assessed them across 32 graphs from four heterogeneous sources. The paper's primary contribution is empirical and descriptive: production data lineage graphs (from one dbt instance and 18 Huawei Cloud DLGs) occupy a distinct region of descriptor space, characterised by high community stability and small normalised spectral gaps, distinct from scientific workflow topologies. This structural characterization is reproducible and does not depend on governance metadata.

As a single-organisation governance pilot at graph-aggregate level, we found a 3-tier rank ordering between D3 algebraic connectivity and documentation rate in one dbt lineage ($\rho = -0.71$, bootstrap CI $[-0.92, -0.27]$). Layer-stratified permutation ($p = 1.0$) showed the relationship is captured entirely by between-layer architecture, not within-layer governance variation. A preliminary cross-organisation comparison at graph-aggregate level on two public dbt projects did not replicate the dbt finding, but construct mismatch between metadata-assigned and folder-derived domains prevents a clean conclusion about replicability.

At node granularity, a centrality-based prediction reproduces a published result. Using the expert-annotated core-asset labels published in [Chen et al., 2024] for six Huawei Cloud lineage graphs, node-level topological features (DATA_FLOW PageRank, ancestor count, out-degree) predict core-asset status under leave-one-graph-out cross-validation at LR AUC $= 0.898 \pm 0.098$ (random baseline 0.546, $p = 0.0003$). The result is robust to excluding shared core-asset IDs (AUC = 0.891). The predictive signal is carried by ordinary node features led by PageRank and out-degree. D1–D4 are not features in this model, so no descriptor-versus-centrality comparison is available. The AUC therefore reproduces the node-centrality finding of Chen et al. [2023] on the same corpus rather than extending it. We report it as such, not as a positive governance finding for the descriptors.

The paper's contribution is therefore: (i) a validated structural characterization framework across 32 lineage graphs from four sources; (ii) an inference protocol for governance–topology studies (permutation inference, bootstrap CIs, null models, subset robustness, layer-stratified permutation, seed robustness, power analysis); (iii) a graph-aggregate governance pilot within one organisation, with an explicit between-layer-architecture caveat; (iv) a node-level core-asset prediction (AUC 0.90) that reproduces the centrality finding of Chen et al. [2023] on the Huawei Cloud DLGs, reported as a reproduction rather than a governance result for the descriptors. The Phase 5 model does not test D1–D4 incremental value. Governance maturity is better measured through process and curated metadata than through graph structure; where structure does carry node-importance signal, simple centrality already captures it.

The descriptors are computable from any lineage graph (dbt manifests, data catalog exports, code-derived dependency graphs) and require no access to the underlying data. The structural characterization component is immediately useful as a comparative tool. The governance prediction component requires further evidence before deployment.

A Reproducibility

All code, anonymized data, and experiment scripts are archived at Zenodo[1] and on GitHub.[2] The repository includes:

- `src/governance_descriptors/`: Python package implementing D1–D4 and statistical utilities (permutation Spearman, Benjamini–Hochberg FDR, partial Spearman correlation).
- `experiments/`: Experiment scripts that reproduce all tables in this paper.
- `data/`: Anonymized dbt lineage data (HMAC-SHA256 hashed node identifiers, domain labels, governance metadata).

Dependencies: Python 3.11+, NetworkX 3.x, NumPy, SciPy, pandas, GUDHI (for persistent homology), scikit-learn (for Experiment 4).

[1] Version DOI 10.5281/zenodo.20209148 for v2.1.0. The concept DOI 10.5281/zenodo.20099999 always resolves to the latest version.

[2] `https://github.com/mhdk1602/multiscale-governance-descriptors`

Anonymization: All node identifiers, domain names, and schema names in the production dbt data were replaced with HMAC-SHA256 hashes prior to export. No organizational identifiers, table names, or column names are present in the released data.

References

Rene Abraham, Johannes Schneider, and Jan vom Brocke. Data governance: A conceptual framework, structured review, and research agenda. *International Journal of Information Management*, 49:424–438, 2019.

Ahmed G.A.H. Ahmed and C. Okan Sakar. DW-Bench: Benchmarking LLMs on data warehouse graph topology reasoning. arXiv preprint arXiv:2604.18964, 2026.

Yunpeng Chen, Ying Zhao, Wenxuan Xie, Yanbo Zhai, Xin Zhao, Jiang Zhang, Jiang Long, and Fangfang Zhou. An empirical study on core data asset identification in data governance. *Big Data and Cognitive Computing*, 7(4):161, 2023. doi: 10.3390/bdcc7040161.

Yunpeng Chen, Ying Zhao, Xuanjing Li, Jiang Zhang, Jiang Long, and Fangfang Zhou. An open dataset of data lineage graphs for data governance research. *Visual Informatics*, 8(1):1–5, 2024. doi: 10.1016/j.visinf.2024.01.001. Data: `https://github.com/csuvis/DataAssetGraphData`.

Samir Chowdhury and Facundo Mémoli. Persistent path homology of directed networks. In *Proceedings of the 29th Annual ACM-SIAM Symposium on Discrete Algorithms (SODA)*, pages 1152–1169, 2018.

Tainã Coleman et al. WfCommons: A framework for enabling scientific workflow research and development. In *Future Generation Computer Systems*, volume 128, pages 16–27, 2022.

DAMA International. *DAMA-DMBOK: Data Management Body of Knowledge.* Technics Publications, 2nd edition, 2017.

Manlio De Domenico and Jacob Biamonte. Spectral entropies as information-theoretic tools for complex network comparison. *Physical Review X*, 6:041062, 2016.

Jean-Charles Delvenne, Sophia N. Yaliraki, and Mauricio Barahona. Stability of graph communities across time scales. *Proceedings of the National Academy of Sciences*, 107:12755–12760, 2010.

Paul J. DiMaggio and Walter W. Powell. The iron cage revisited: Institutional isomorphism and collective rationality in organizational fields. *American Sociological Review*, 48(2):147–160, 1983.

Miroslav Fiedler. Algebraic connectivity of graphs. *Czechoslovak Mathematical Journal*, 23(98): 298–305, 1973.

Yuxuan Hou et al. Enhanced error fault localization framework based on TCN and GCN for data pipelines. *Scientific Reports*, 16:4430, 2026.

Shenyang Huang, Yasmeen Hou, Bryan Hooi, Bin Wu, Christopher Jin, and Reihaneh Rabbany. Laplacian change point detection for dynamic graphs. In *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 349–357, 2020. doi: 10.1145/3394486.3403077.

Shenyang Huang, Samy Coulaud, Reihaneh Rabbany, and Tsuyoshi Murata. Laplacian change point detection for single and multi-view dynamic graphs. *ACM Transactions on Knowledge Discovery from Data*, 18(3), 2024. doi: 10.1145/3631609.

Yasutaka Kamei, Emad Shihab, Bram Adams, Ahmed E. Hassan, Audris Mockus, Anand Sinha, and Naoyasu Ubayashi. A large-scale empirical study of just-in-time quality assurance. *IEEE Transactions on Software Engineering*, 39(6):757–773, 2013. doi: 10.1109/TSE.2012.70.

Vijay Khatri and Carol V. Brown. Designing data governance. *Communications of the ACM*, 53(1):148–152, 2010.

Danai Koutra, Neil Shah, Joshua T. Vogelstein, Brian Gallagher, and Christos Faloutsos. DeltaCon: Principled massive-graph similarity function with attribution. *ACM Transactions on Knowledge Discovery from Data*, 10(3):28:1–28:43, 2016. doi: 10.1145/2824443.

Renaud Lambiotte, Jean-Charles Delvenne, and Mauricio Barahona. Random walks, Markov processes and the multiscale modular organization of complex networks. *IEEE Transactions on Network Science and Engineering*, 1(2):76–90, 2014.

Dineshkumar Malempati Hari. Boundary-risk detection for master data governance. *ACM Journal of Data and Information Quality*, 2026a. Under review.

Dineshkumar Malempati Hari. Box-covering descriptors for enterprise data graphs. Unpublished phase 0 feasibility analysis, no public artifact released. The finding it produced is restated in full where it is cited, 2026b.

Dineshkumar Malempati Hari. Static and temporal fractal coupling between volatility and trading volume: Evidence from S&P 500 stocks, 2015–2026. Zenodo preprint, 2026c.

Albert Meroño Peñuela et al. Knowledge graphs as backbone of data governance in AI. *Journal of Web Semantics*, 2024.

Nina Otter, Mason A. Porter, Ulrike Tillmann, Peter Grindrod, and Heather A. Harrington. A roadmap for the computation of persistent homology. *EPJ Data Science*, 6:17, 2017.

Giovanni Petri, Martina Scolamiero, Irene Donato, and Francesco Vaccarino. Topological strata of weighted complex networks. *PLOS ONE*, 8(6):e66506, 2013.

Jörg Reichardt and Stefan Bornholdt. Statistical mechanics of community detection. *Physical Review E*, 74:016110, 2006.

W. Richard Scott. *Institutions and Organizations: Ideas, Interests, and Identities.* SAGE Publications, 4th edition, 2013.

Seokyong Seo et al. Structural quality metrics for knowledge graphs. *Semantic Web Journal*, 2022.

Vincent A. Traag, Ludo Waltman, and Nees Jan van Eck. From Louvain to Leiden: Guaranteeing well-connected communities. *Scientific Reports*, 9:5233, 2019.

Kristin Weber, Boris Otto, and Hubert Osterle. One size does not fit all—a contingency approach to data governance. *Journal of Data and Information Quality*, 1(1):1–27, 2009.